

Machine learning LAB 1

Name :Karthik R Nayak

SRN: PES2UG24CS217

DATE: 6-8-2026

ANALYSIS REPORT

Investigation 6

Insight 1: Complete and Cancelled Booking Status

From the data, it is evident that there are many complete bookings compared to cancelled bookings. In this case, more than two-thirds of the bookings were successful, whereas one-third of the bookings have been cancelled.

Business Importance:

Despite the fact that the hotel is able to maintain many bookings, there is a significant percentage of cancellations which may have some negative impacts on planning.

Insight 2: Urban Hotels Receive More Bookings

In terms of hotel types, there is a higher amount of bookings for Urban Hotels as opposed to Luxury Resorts.

Reason:

One reason as to why this happens is because urban hotels are mostly favored by business travelers as well as tourists due to their location.

Business Importance:

More operational capacity is needed for Urban Hotels due to higher bookings.

Insight 3: Seasonal Booking Demand

Seasons also determine the number of bookings made. This is the case whereby summer has the most bookings, spring comes second, while winter has the least demand.

Business Importance:

This helps the hotel understand seasonal demand.

Insight 4: Busiest Booking Months

Based on the monthly booking chart, the hotel's peak months for bookings are July and August, whereas the low months are January and December.

Business Relevance:

Hotel management needs to be ready for high occupancy in peak months with adequate staffing and customers' support. In low demand months, special offers can be launched to increase occupancy rates.

Insight 5: Booking Channels Usage

According to the booking channels chart, the greatest number of bookings is made through the Travel Agent/Tour Operator (TA/TO) channel, while direct and corporate bookings make up a relatively small portion.

Business Relevance:

Travel agencies continue to be the most efficient sales channel for the hotel. It is important to maintain a good relationship with partners in order to keep bookings.

Insight 6: Customer Type

Customer category analysis shows that the greatest share of bookings belongs to Regular customers, while Corporate Contracts and Group Bookings are minimal.

Business Relevance:

The marketing campaign should focus mainly on individual travelers and at the same time explore new markets within the corporate and group segment.

Insight 7: Pricing Behaviour

The Average Daily Rate (ADR) distribution suggests that the majority of the room price falls within the ₹70 – ₹120 range.

Business Significance:

The pricing strategy of the hotel lies in the mid-level, implying that most customers like mid-range rooms.

Insight 8: Revenue by Booking Segments

The Average Daily Rate (ADR) analysis by booking segments suggests that both Online Travel Agencies and Direct bookings have higher ADR, while the Corporate and Undefined segments have lower ADR.

Business Significance:

The hotel should focus on booking segments having higher value while analysing other lower performing market segments in terms of their pricing strategy.

Insight 9: Cancellation by Type of Hotel

Based on the analysis of cancellation, Urban Hotels get more cancellation than Luxury Resorts. But Urban Hotels also get more reservations compared to Luxury Resorts.

Business Significance:

Hotel Management needs to find out the reason for cancellation in the Urban Hotel and try to put some steps like confirmation of bookings, flexibility in price, or deposit before booking in order to avoid revenue loss.

Insight 10: Relationship among Business Variables

Correlation analysis reveals:

Number of weekday nights shows a very strong positive relationship with total stay night (≈ 0.95). Number of weekend nights also has a strong contribution towards total stay (≈ 0.77). Number of Family shows a strong relationship with the number of children (≈ 0.89). Number of Adults shows a strong positive relationship with Total guests (≈ 0.78).

Most of the other variables show weak relationships, which means hotel bookings depend on many independent variables.

Business Importance:

These relationships help management understand customer behaviour and improve room allocation, family-oriented services, and operational planning.

Investigation 7 – Recommendations for the Business

1. Decrease Cancellation Rate of Bookings

In order to decrease the rate of cancellation of bookings, use reminders and rescheduling possibilities as well as partial advance payment, especially concerning Urban Hotels.

2. Improve the Functioning of Urban Hotel

As the Urban Hotels get most bookings, employ more staff for housekeeping and customer services during the peak demand time.

3. Implement the Seasonal Pricing Strategy

Increase the price of the room during the high-demand time period, namely Summer, July, and August, and implement the discount pricing strategy during Winter and other low-demand periods.

4. Build Good Relationships with the Travel Agencies

As the TA/TO bookings make the biggest percentage of bookings, keep cooperating with the travel agencies using an attractive commission scheme.

5. Increase Corporate and Group Bookings

Design the special corporate package and facilities for conference in order to attract more bookings of underrepresented customer segments.

6. Apply the Revenue Management Method

Watch out the ADR of all booking segments and apply the appropriate pricing strategy in order to maximize the revenue of your company.

7. Customer Retention

Design the loyalty program and offers for loyal customers.

8. Create a seasonal planning

Using historical booking trends try to predict occupancy, staff scheduling, room availability, and inventory requirements throughout the year.